

Session 9 – Near Things More Related Than Distance Things in London
(Spatial Dependence, Spatial Regression and Surface Analysis
with GeoDa and QGIS)

Objectives: To analyse spatial dependence / correlation for geographic profiling and reasoning. To analyse spatial regression for analytic prediction modelling. To interpolate point events and analyse raster data with surface techniques.

Task 1: Spatial Dependence / Correlation of Crimes and Deprivation in London

Problem solving: Measuring, mapping and analysing spatial dependence / correlation of crimes and deprivation to support policing strategies

Functionality: Moran's I in GeoDa

Data set: Crime variables and Multiple Index of Deprivation by London Borough

Input the Shape file of "london_life_polygon.shp" in GeoDa, and create a weights file for calculating Moran's I.

GeoDa -> Tools -> Weights -> Create (*suggest to select Rook Contiguity*)

On "london_life_polygon.shp", measure spatial dependence with following functionalities by variables of 'DEPRIV' (score of deprivation), 'CRIME_1000' (total counts of crime per thousand persons), 'BURGLARY' (counts of burglary per thousand persons), 'CRIM_DAMAG' (counts of crime damage per thousand persons), 'DRUG_OFFEN' (counts of drug offence per thousand persons), 'VEHICLES' (counts of vehicle theft per thousand persons), 'VIOLENCE' (counts of violence crime per thousand persons).

GeoDa -> Space -> Univariate Moran's I

-> Univariate Local Moran's I

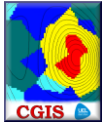
Does deprivation has strong spatial dependence? Which type of crime has highest spatial dependence? What does the Univariate Moran's I imply for deprivation, total counts of crime and counts of burglary at the geographic scale of borough? Copy Moran Scatter Plot and Cluster Map of 'DEPRIV', 'CRIME_1000' and 'BURGLARY' to your portfolio.

On "london_life_polygon.shp", measure spatial correlation with following functionalities by the pair of 'DEPRIV' & 'CRIME_1000' and the pair of 'DEPRIV' & 'BURGLARY'.

GeoDa -> Space -> Bivariate Moran's I

-> Bivariate Local Moran's I

What does the Bivariate Moran's I imply for the pair of 'DEPRIV' and 'CRIME_1000' and the pair of 'DEPRIV' and 'BURGLARY' at the geographic scale of borough? Copy Moran Scatter Plots and Cluster Maps to your portfolio.



On "london_life_polygon.shp", measure smoothed (or adjusted) spatial dependence with following functionalities by rates of 'BURGLARY' (event variable) / 'CRIME_1000' (base variable), and 'DRUG_OFFEN' (event variable) / 'CRIME_1000' (base variable).

GeoDa -> Space -> Moran's I with EB Rate

-> Local Moran's I with EB Rate

Can you see any difference in results for 'BURGLARY' and 'DRUG_OFFEN' from Moran's I with EB Rate and Univariate Moran's I?

Try "london_ward_met_per1000.shp" with Univariate Local Moran's I. by variables of 'Total' (total counts of crime per thousand persons) and 'Burglary' (counts of crime burglary per thousand persons).

Find out any difference in spatial dependence between borough-level and ward-level caused by MAUP. Copy Moran Scatter Plots and Cluster Maps to your portfolio.

Try "london_pop_time_plg.shp", measure spatial-temporal dependence with following functionalities by the population estimates in year1 and year6.

GeoDa -> Space -> Differential Moran's I

-> Differential Local Moran's I

What does the Differential Moran's I imply for temporal change of population at the geographic scale of borough? Copy Moran Scatter Plots and Cluster Maps to your portfolio.

Task 2: Spatial Regression of Health Indicators in London

Problem solving: Modelling spatial Regression of health indicators to support healthcare strategies

Functionality: Regression in GeoDa

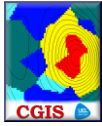
Data set: Health variables and Multiple Index of Deprivation by London Borough

On "london_life_polygon.shp", run regression modelling with following functionality by 'LIFE_MALE' (life expectancy of male residents) as Dependent Variable and 'DEPRIV' (score of deprivation), 'P_SMOKE' (population percentage of smokers), 'P_BINGE' (population percentage of binge drinkers), 'P_OBESE' (population percentage of obesity) and 'P_HEALTHY' (population percentage of healthy residents) as Covariates. Please try both Classic and Spatial Lag models.

GeoDa -> Regression

Copy regression reports to your portfolio.

Try Session9_regression.r and copy results of regression modelling to your portfolio. Please compare results from GeoDa and R.



Task 3: Population Surface in Haringey

Problem solving: Develop surfaces of social variables with interpolation to compare or integrate data at different spatial scales

Functionality: interpolation in QGIS

Data set: Population centroids of LSOA in Haringey

Add "Population_Haringey_LSOA_PWC.shp" and "haringey_boundary.shp". Click the button of "Zoom Full".

QGIS -> Processing -> Toolbox > search "interpolation"
choose "IDW interpolation"

(In the window of Interpolation; Choose "Distance Coefficient" as 4; Zoom to Layer of "Population_Haringey_LSOA_PWC", Use Map Canvas Extent, suggested numbers of rows as 300)

QGIS -> Raster -> Extraction -> Clipper Raster by Mask Layer

QGIS -> Layer -> Layer Properties -> Symbolology

(right click of mouse -> Properties -> Symbolology -> Render type -> Singleband pseudocolor, then change the legend)

QGIS -> Raster -> Extraction -> Contour
(try interval of contour lines as 100)

Try interpolation with different methods and parameters. Is interpolation sensitive to methods and parameter settings? (just try, don't need to copy to portfolio)

Task 4: Terrain Surface around Waterloo Bridge

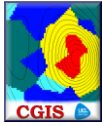
Problem solving: Explore natural surfaces with remote sensing techniques to provide information for spatial data analysis

Functionality: Terrain Analysis in QGIS

Data set: LiDA DSM around Waterloo Bridge

Add "TQ3080_DSM_2M.asc", Analysis the DSM with following functionalities:

QGIS -> Processing -> Toolbox > search "raster terrain analysis"
under "Raster terrain analysis"
choose "Ruggedness index"
choose "Slope"
choose "Aspect"



choose "Hillshade"

choose "Relief"

Please export specified Scatter Plots and Cluster Maps from **Task 1** (three variables for Univariate Local Moran's I, one pair of variables for Bivariate Local Moran's I, one variable for Local Moran's I with EB Rate, one variable for "london_ward_met_per1000.shp" with Univariate Local Moran's I, the population estimates in year1 and year6 for Differential Local Moran's I), and put them into your portfolio.

Please copy results of regression modelling from **Task 2**, export the interpolated raster image from **Task 3**, and put them into your portfolio.

(QGIS: Project -> Import/Export -> Export Map to Image, GeoDa: right click of you mouse -> Save Image as)